AN OPEN IMMERSION IS OF FINITE PRESENTATION EXACTLY WHEN IT IS QUASI-COMPACT

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- An open immersion is locally of finite presentation, cited from the same chapter. Cited at proof lines 4198-4200. Provides the local half of the finite presentation condition.
- An immersion is separated and hence quasi-separated, cited from the schemes chapter. Cited at proof lines 4200-4202. Provides the quasi-separated half for free, reducing the remaining condition to quasi-compactness.
- The definition of a morphism of finite presentation as one that is locally of finite presentation, quasi-compact and quasi-separated. Cited at proof lines 4202-4204. Combines the three properties into the claimed equivalence.

Source reference. Original definition 01TP: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2382–2382. Same-chapter reference used in the retained author text at source line(s) 4203. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 01TT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2386–2386. Same-chapter reference used in the retained author text at source line(s) 4199. It is supplied only as a navigation pointer

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Source reference. Original lemma 01L7: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2076–2076. Same-chapter reference used in the retained author text at source line(s) 206, 260, 4201, 9139, 9942, 12927. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Lemma 0.1. *Any open immersion is of finite presentation if and only if it is quasi-compact.*

Proof. We have seen (Lemma 01TT) that an open immersion is locally of finite presentation. We have seen (Schemes, Lemma 01L7) that an immersion is separated and hence quasi-separated. From this and Definition 01TP the lemma follows. □